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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

Supplementary Examination - December 2025

EC3003-1 – DIGITAL SIGNAL PROCESSING

(FOR EC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- Given $x(n)$ as a finite length sequence with DFT $X(k) = (0, 1 + j, 1, 1 - j)$. If $x_3(n) = x((n - 1)_4)$ then $X_3(k)$ is
 - $(0, 1 - j, -1, 1 + j)$
 - $(0, 1 + j, -1, 1 - j)$
 - $(1 - j, 0, -1, 1 + j)$
 - $(1 - j, 1, -1, 1 + j)$
- The circular convolution of two sequences in time domain is equivalent to
 - Multiplication of DFTs of two sequences
 - Summation of DFTs of two sequences
 - Difference of DFTs of two sequences
 - Square of multiplication of DFTs of two sequences
- Which of the following is true regarding the number of computations required to compute an N point DFT?
 - N^2 complex multiplications and $N(N+1)$ complex additions
 - N^2 complex multiplications and $N(N-1)$ complex additions
 - N^2 complex additions and $N(N+1)$ complex multiplications
 - N^2 complex additions and $N(N-1)$ complex multiplications
- If $W_4^{100} = W_X^{200}$, then what is the value of x ?
 - 2
 - 4
 - 8
 - 16
- The overlap save method is used to calculate
 - The discrete convolution between a sampled signal and a finite impulse response (FIR) filter
 - The discrete convolution between a sampled signal and an infinite impulse response (IIR) filter
 - The discrete convolution between a very long signal and a finite impulse response (FIR) filter
 - The discrete convolution between a very long signal and an infinite impulse response (IIR) filter
- Radix - 2 FFT algorithm performs the computation of DFT in
 - $(N/2) \log_2 N$ multiplications and $2 \log_2 N$ additions
 - $(N/2) \log_2 N$ multiplications and $N \log_2 N$ additions
 - $\log_2 N$ multiplications and $(N/2) \log_2 N$ additions
 - $N \log_2 N$ multiplications and $(N/2) \log_2 N$ additions
- DIT algorithm divides the sequence into
 - Positive and negative values
 - Even and odd samples
 - Upper higher and lower spectrum
 - Small and large samples

8. **Overlap-Add Method deals with the principle**
- A) The linear convolution of a discrete-time signal of length L and a discrete-time signal of length M produces a discrete-time convolved result of length $L + M - 1$
- B) The linear convolution of a discrete-time signal of length L and a discrete-time signal of length M produces a discrete-time convolved result of length $L + M$
- C) The linear convolution of a discrete-time signal of length L and a discrete-time signal of length M produces a discrete-time convolved result of length $2L + M - 1$
- D) The linear convolution of a discrete-time signal of length L and a discrete-time signal of length M produces a discrete-time convolved result of length $2L + 2M - 1$
9. **The IIR filter transfer function contains**
- A) Poles only
- B) Zeros only
- C) Data is inadequate
- D) Poles and Zeros
10. **Impulse Invariance Transform:**
- A) Transforms the entire left half of s -plane in to the unit circle in the z -plane
- B) Transforms a strip on the left half of s -plane between $-\pi$ to π
- C) Transforms a strip on the right half of s -plane between $-\pi$ to π
- D) Transforms the entire right half of s -plane in to the unit circle in the z -plane
11. **The number of delay elements to be used to implement the transfer function $H(z)$ in DF-I, given**
- $$H(z) = \frac{1 - 0.3z^{-1} + 0.4z^{-2}}{1 - 0.25z^{-1} + 0.46z^{-3}}$$
- is
- A) 4
- B) 2
- C) 5
- D) 3
12. **The order of a Low Pass Butterworth filter, given that the 3 dB bandwidth is 500Hz and 40 dB attenuation is at 1000Hz is**
- A) 7
- B) 5
- C) 9
- D) 4
13. **The realization of FIR filter by frequency sampling realization can be viewed as cascade of how many filters?**
- A) Two
- B) Three
- C) Four
- D) Five
14. **If a three-stage lattice filter with coefficients $K_1=1/4$, $K_2=1/2$, $K_3=1/3$, then what are the FIR filter coefficients for the direct form structure?**
- A) $(1, 8/24, 5/8, 1/3)$
- B) $(1, 5/8, 13/24, 1/3)$
- C) $(1/4, 13/24, 5/8, 1/3)$
- D) $(1, 13/24, 5/8, 1/3)$
15. **What is the width of the main lobe of the frequency response of a rectangular window of length $M-1$?**
- A) π/M
- B) $2\pi/M$
- C) $4\pi/M$
- D) $8\pi/M$
16. **The basic building blocks essential to carry out DSP computations will not include this**
- A) Multiplier
- B) Shifter
- C) ALU
- D) USB
17. **Limiting the accumulator contents to its maximum limits is achieved by a simple logic circuit known as**
- A) Accumulation Logic Circuit
- B) Address Generation Unit
- C) Saturation Logic Circuit
- D) Program Sequencer
18. **Fixed point filter implementation uses**
- A) 2's complement integer format
- B) Q point format
- C) Single precision format
- D) Double precision format
19. **The following addressing mode is used in FIR filter implementation**
- A) Bit reversed addressing mode
- B) Immediate addressing mode
- C) Circular addressing mode
- D) Direct addressing mode
20. **The architecture that has two memories**
- A) Von Neumann architecture
- B) Harvard Architecture
- C) Modified Harvard Architecture
- D) VLIW Architecture

PART - B: DESCRIPTIVE ANSWER QUESTIONS**Unit – I**

	Marks	BT*	CO*	PO*
1. a) Find the N- point DFT of the sequence $x(n) = \cos(2\pi k_0 n/N) \quad 0 \leq n \leq N-1$	4	L*3	1	1
b) The 8-point DFT of a length- 8 sequence $x(n)$ is given by, $X(k) = \{2, 1+j, 2-j, 3+j, 1, 3-j, 2+j, 1-j\}$ Evaluate the function $\sum_{n=0}^7 e^{j\pi n} x(n)$	4	L3	1	1
c) Find the sequence $x(n)$ corresponding to the 8-point DFT using DIT-FFT algorithm. $X(k) = \{4, 1-j2.414, 0, 1-j0.414, 0, 1+j0.414, 0, 1+j2.414\}$	8	L3	2	1
2. a) Compute 4-point circular convolution of the sequences given by $x(n) = \{1, 0, 5\}$ and $h(n) = \{0, 5, 1\}$ using DFT and IDFT method (i.e., Stockham's method)	4	L3	1	1
b) Find the 4-point DFT of the sequence $x(n) = \{2+j2, -3+j1, 2+j1, -1+j3\}$ and find 4-point DFT of $y(n)$ using appropriate symmetry properties $y(n) = \{1, -1.5, 1, -0.5\}$	4	L3	1	1
c) Find the output $y(n)$ of a filter whose impulse response $h(n) = \{1, 1, 1\}$ and input signal $x(n) = \{3, -1, 0, 1, 3, 2, 0, 1, 2, 1\}$ using overlap-save method. Assume 5 point circular convolution only.	8	L3	2	1
3. a) The 8-point DFT of the following sequence is given by $X(k)$ $x(n) = \begin{cases} 1 & ; 0 \leq n \leq 3 \\ 0 & ; 4 \leq n \leq 7 \end{cases}$ Compute the DFT of the sequence $x_1(n) = \{0, 0, 1, 1, 1, 1, 0, 0\}$ using Circular Time Shift Property.	4	L3	1	1
b) Find 4-point DFTs of the sequences $x_1(n) = \{1, 1, 2, 1\}$ and $x_2(n) = \{1, 0, 1, 0\}$ and thereby find the 4-point DFT of the sequence $x_3(n) = \{4, 0, 8, 0\}$	4	L3	1	1
c) Given $x(n) = n+1$ for $0 \leq n \leq 7$. Find $X(k)$ using DIF-FFT algorithm.	8	L3	2	1

Unit – II

4. a) Design a digital Butterworth IIR low pass filter with 3 dB frequency of 10 rad/sec and attenuation of atleast 40 dB for frequencies larger than 100 rad/sec. The sampling frequency is 1000 Hz. Use impulse invariance transformation.	8	L3	3	1
b) Design a lowpass FIR filter using rectangular window with $M = 7$ and with a cutoff frequency of 1 rad/sec.	8	L3	4	1
5. a) Design a Type-1 Chebyshev lowpass filter with ripple of 1 dB in the passband $0 \leq \omega \leq 0.2\pi$ and attenuation of atleast 15 dB in the stopband $0.3 \leq \omega \leq \pi$. Use impulse invariance method.	8	L3	3	1
b) Determine the coefficients $h(n)$ for a linear phase FIR filter of length $M=15$ which has symmetric unit impulse response and the frequency response that satisfies the conditions $H\left(\frac{2\pi}{15}k\right) = \begin{cases} 1 & ; k = 0, 1, 2, 3 \\ 0 & ; k = 4, 5, 6, 7 \end{cases}$	8	L3	4	11

6. a) Design a Type-1 Chebyshev lowpass filter to meet the following specifications.

- (a) passband ripple ≤ 2 dB
 (b) passband edge frequency = 1 rad/sec
 (c) stop band attenuation ≥ 20 dB
 (d) stop band edge frequency = 1.3 rad/sec

8 L3 3 1

- b) A FIR filter is given by $y(n) = x(n) + (2/5)x(n-1) + (3/4)x(n-2) + (1/3)x(n-3)$. Draw the lattice structure.

8 L3 4 1

Unit – III

7. a) Discuss the following DSP Architectural Units with necessary diagrams.

- (i) Braun Multiplier
 (ii) Barrel Shifter

8 L2 5 1

- b) It is required to find the sum of 64, 16-bit numbers.

- i) How many bits should the accumulator have so that the sum can be computed without the occurrence of overflow error or loss of accuracy?
 ii) It is decided to have an accumulator with only 16 bits but shift the numbers before the addition to prevent overflow, by how many bits should each number be shifted?
 iii) If all the numbers are fixed point integers, what is the actual sum of the numbers?
 iv) If a sum of 256 products is to be computed using a pipelined MAC unit, and if the MAC execution time of the unit is 100nsec, what will be the total time required to complete the operation?

8 L3 5 1

8. a) Discuss the following DSP Architectural Units with necessary diagrams.

- (i) MAC unit
 (ii) ALU

8 L2 5 1

- b) (i) Convert into Q15 notation $N=+0.5$
 (ii) Find the signed Q15 representation for the decimal number 0.560123.
 (iii) Add the two numbers 0.100011110110010 and 1.110101110000010
 (iv) Convert the Q-15 signed number 0.100011110110010 to the decimal number.

8 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
